

A-1 LAUNCHPAD 2026 · ROUND 2 · SOFTWARE DEVELOPMENT / AI-ML

Smart VMS

AI Intrusion Detection & False-Alarm Suppression

A video management system smart enough to know a swaying tree isn't an intruder.

Case Study: Smart Video Management System (VMS)

Team: Byte Breakers

College: National Institute of Technology Delhi

Smart VMS

AI Intrusion Detection & False-Alarm Suppression · Team Byte Breakers · NIT Delhi

The pain

Traditional surveillance systems force operators to switch between live monitoring and recorded playback during an investigation — costing precious seconds during a real incident. Worse, naïve motion detection floods operators with false alarms (wind, animals, foliage), causing alert fatigue: the one real intrusion looks identical to the hundred false ones before it.

Our one-line solution

A unified VMS where live feeds, recorded playback, and AI-flagged events live on one interactive timeline — with a four-layer filtering pipeline smart enough to know a swaying tree isn't an intruder.

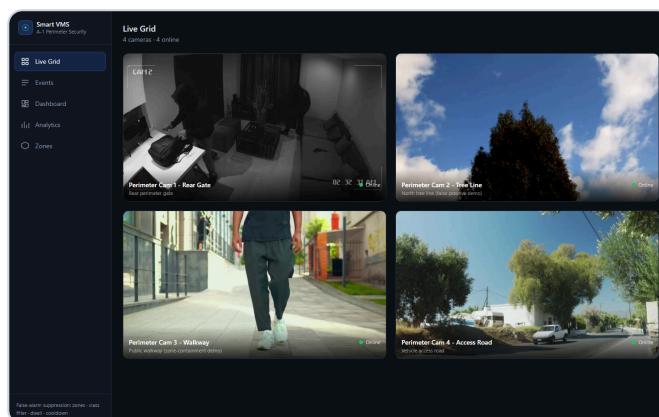
Architecture

```

Sample clips (looped, 4 cameras)
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  v
FastAPI backend -- MJPEG stream, H.264 segment recording
  |
  v
Detection pipeline -- MOG2 motion gate -> YOLOv8 classify -> zone test -> dwell/cooldown
  |
  v
SQLite + clips/thumbnails on disk
  |
  v
React frontend -- live grid . timeline . dashboard . zone editor . analytics
  ^
  +-- WebSocket real-time push
  
```

Tech stack

Python 3.13 + FastAPI · OpenCV + MOG2 · Ultralytics YOLOv8n · SQLite · React + Vite + Tailwind CSS · Recharts · WebSocket



Live Grid — all 4 camera feeds, online status, dark operator UI

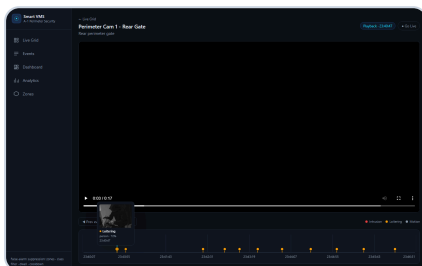
The four-layer false-alarm pipeline

Layer	What it does	Why it matters
1. Motion pre-filter (OpenCV MOG2)	Cheap background subtraction gates the expensive model	YOLO only runs on frames where something actually moved
2. Object-class filter (YOLOv8)	Must classify as person / car / truck / bus / bicycle / motorcycle	Leaves, shadows, and camera noise discarded before a zone is ever checked
3. Zone containment	Detection centroid must fall inside an enabled, operator-drawn polygon	A person on a public path outside the fence doesn't alert; crossing the perimeter does
4. Dwell + cooldown	Escalation requires N seconds of continued presence; a 30s cooldown blocks duplicate re-alerts	One lingering intruder generates one alert, not fifty

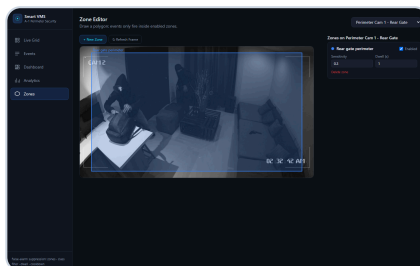
Verified result: 878 raw motion triggers → 78 confirmed events = **91% false-alarm reduction**, with the wind/foliage test camera sitting at zero detections throughout.

Core features

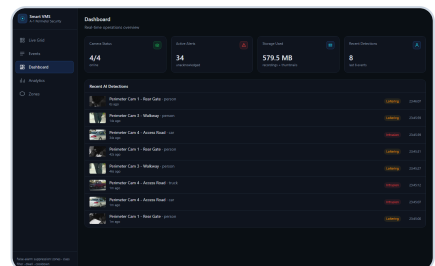
- Live multi-camera grid with real-time alert flashing
- Interactive canvas timeline — colored event markers, hover-to-preview thumbnail, one-click seek to any event's exact footage
- Severity-coded bounding boxes on event thumbnails — red for intrusion, orange for loitering
- Zone editor — draw a polygon on a live paused frame, tune sensitivity and dwell time per zone, no restart required
- Dashboard — camera status, active alerts, storage usage, recent detections, all live via WebSocket
- Analytics — false-alarm reduction stat, detections per hour, detections per zone



Timeline — click-to-seek



Zone editor



Dashboard

Impact, in numbers

91%

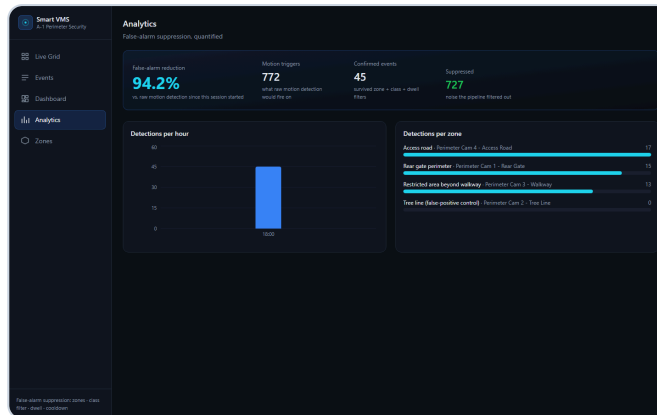
FALSE-ALARM REDUCTION

<3s

ALERT TO FOOTAGE

1 click

MARKER TO PLAYBACK



Analytics — false-alarm reduction, computed live, not projected

What's built (all of it, verified live — not mocked)

Live grid, continuous H.264 recording, full 4-layer detection pipeline, event generation with severity-colored bounding-box thumbnails, click-to-seek timeline, zone editor, dashboard, analytics — every required capability plus both win-booster differentiators. CPU-only detection, no GPU required. Tested end-to-end on both the sample clips and a real live webcam feed.

Roadmap (deliberately not built for this prototype)

Real RTSP/ONVIF camera integration, cloud storage & retention, user authentication/roles/audit trails, face/license-plate recognition, native mobile apps, direct integration with A-1's Vigil PIDS sensor network, edge deployment.

Relevance to A-1

This directly mirrors the false-alarm problem A-1's PIDS product line already faces in the field — the same zone + class + dwell filtering logic applies whether the sensor is a camera or a fence vibration sensor.

Links

GitHub: <https://github.com/Priyanshu-44/VMS-application>

Demo video: <https://drive.google.com/file/d/1P0PAuc7I6PgWUa2UF9chmtmrAyLpWdFZ/view?usp=sharing>

Resume — Team Member 1: <https://drive.google.com/file/d/1aHB2G16ht81A0CwcPx0MkCuKXVSJs4V/view?usp=sharing>

Resume — Team Member 2: https://drive.google.com/file/d/1WroLWcwPpYsCe8TEplbXUQSi5SL-CN9j/view?usp=drive_link